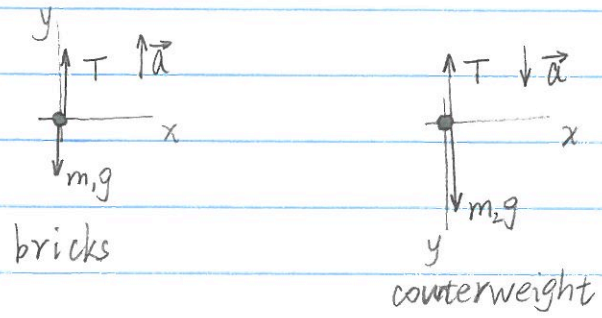
Phys. 2110 HW # 5 Solution

5.19, 55, 62, 65, 81, 82, 90, 99, 115

5.19 (a)



(b) Apply $\Sigma F_y = m a_y$ to each object. The acceleration magnitude is the same for the two objects.

For the bricks take $+y$ to be upward since $\vec{a}$ for the bricks is upward. For the counterweight take $+y$ to be downward since $\vec{a}$ is downward.

bricks: $\Sigma F_y = m a_y \qquad T - m_1 g = m_1 a$

counterweight: $\Sigma F_y = m a_y \qquad m_2 g - T = m_2 a$

Add these two equations to eliminate T

$$(m_2 - m_1) g = (m_1 + m_2) a$$

$$a = \frac{m_2 - m_1}{m_2 + m_1} g = \frac{28.0\,kg - 15.0\,kg}{28.0\,kg + 15.0\,kg} \times 9.80\,m/s^2 = 2.96\,m/s^2$$

(c) $T - m_1 g = m_1 a$ gives $T = m_1 (a + g) = 15.0\,kg \times (2.96\,m/s^2 + 9.80\,m/s^2)$

$$= 191\ N$$

The tension is 1.30 times the weight of the bricks; this causes the bricks to accelerate upward. The tension is 0.696 times the weight of the counterweight; this causes the counterweight to accelerate downward.

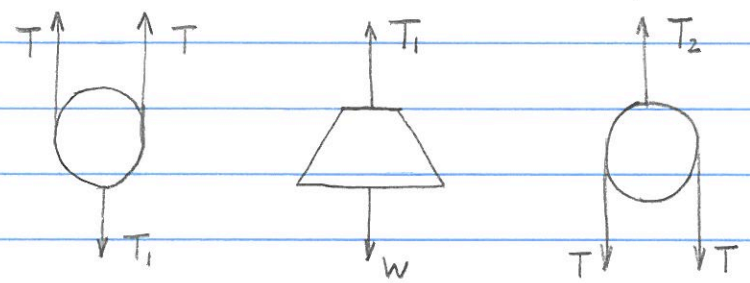
5.55 (a) Setting $a_{rad} = g$ and solving for the period T gives

$$T = 2\pi\sqrt{\frac{R}{g}} = 2\pi\sqrt{\frac{400m}{9.80m/s^2}} = 40.1s$$

so the number of revolutions per minute is $(60s/min) \times 40.1s = 1.5$ rev/min

(b) The lower acceleration corresponds to a longer period, and hence a lower rotation rate, by a factor of the square root of the ratio of the acceleration, $(1.5 \text{ rev/min}) \times \sqrt{\frac{3.70}{9.80}} = 0.92$ rev/min

5.62



The tension in the lower chain balances the weight and so is equal to W. The lower pulley must have no net force on it, so twice the tension in the rope must be equal to W and the tension in the rope, which equals F, is $\frac{W}{2}$. Then, the downward force on the upper pulley due to the rope is also W, and so the upper chain exerts a force W on the upper pulley, and the tension in the upper chain is also W. $T_1 = W$, $T = \frac{W}{2}$, $T_2 = W$

5.65 (a) The tension in the cord must be $m_2 g$ in order that the hanging block move at constant speed. This tension must overcome friction and the component of the gravitational force along the incline, so

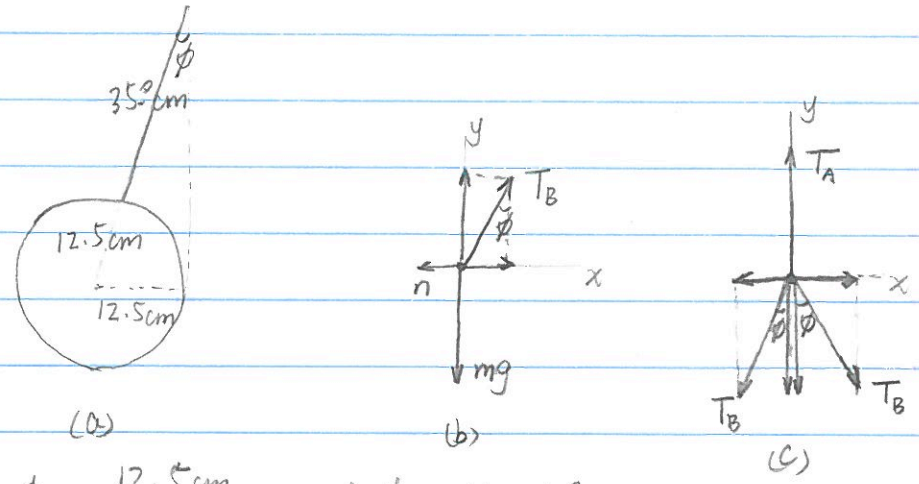
(2) $m_2 g = m_1 g \sin\alpha + \mu_k m_1 g \cos\alpha$, and $m_2 = m_1 \sin\alpha + \mu_k m_1 \cos\alpha$

(b) In this case, the friction force acts in the same direction as the tension on the block of mass m_1, so

$m_2 g = m_1 g \sin\alpha - \mu_k m_1 g \cos\alpha$, and $m_2 = m_1 \sin\alpha - \mu_k m_1 \cos\alpha$

(c) Similar to the analysis of parts (a) and (b), the largest m_2 is $m_1 \sin\alpha + \mu_s m_1 \cos\alpha$, and the smallest m_2 is $m_1 \sin\alpha - \mu_s m_1 \cos\alpha$.

5.81



(a) $\sin\phi = \dfrac{12.5\text{ cm}}{47.5\text{ cm}}$ and $\phi = 15.26°$

$\Sigma F_y = ma_y$ applied to the ball gives $T_B \cos\phi - mg = 0$

$$T_B = \frac{mg}{\cos\phi} = \frac{(15.0\text{ kg})(9.80\text{ m/s}^2)}{\cos 15.26°} = 152\text{ N}$$

Then $\Sigma F_y = m a_y$ applied in Figure (c) gives $T_A - 2T_B \cos\phi = 0$

$T_A = 2 \times (152\text{ N}) \cos\phi = 294\text{ N}$

(b) $\Sigma F_x = m a_x$ applied to the ball gives $n - T_B \sin\phi = 0$, and

$n = (152\text{ N}) \sin\phi = 40.0\text{ N}$

5.82 The maximum acceleration possible due to static friction is $(0.19)(9.80\text{ m/s}^2) = 1.86\text{ m/s}^2$, and so the block will move relative to the truck; the acceleration of the box would be $\mu_k g = (0.15) \times (9.80\text{ m/s}^2) = 1.47\text{ m/s}^2$. The difference between the distance the truck moves and the distance the box moves will be 1.80 m

after a time

$$t = \sqrt{\frac{2\,\Delta x}{a_{truck} - a_{box}}} = \sqrt{\frac{2\,(1.80\,m)}{(2.20\,m/s^2 - 1.47\,m/s^2)}} = 2.221\,s$$

In this time, the truck moves $\frac{1}{2} a_{truck} t^2 = \frac{1}{2}(2.20\,m/s^2)(2.221\,s)^2$

$= 5.43\,m$

5.90 The floor exerts an upward force n on the box, obtained from $n - mg = ma$, $n = m(a+g)$. The friction force that needs to be balanced is

$$\mu_k n = \mu_k m(a+g) = (0.32)(28.0\,kg)(1.90\,m/s^2 + 9.80\,m/s^2) = 105\,N$$

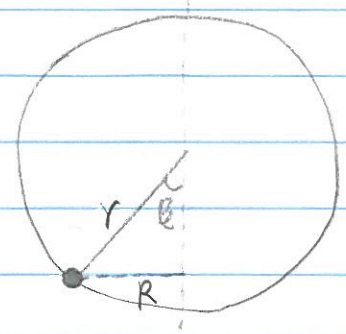
5.99 (a) For the monkey to move up, $T > mg$. The bananas also move up.

(b) The bananas and monkey move with the same acceleration and the distance between them remains constant.

(c) Both the monkey and bananas are in free fall. They have the same initial velocity and as they fall the distance between them doesn't change.

(d) The bananas will slow down at the same rate as the monkey. If the monkey comes to a stop, so will the bananas.

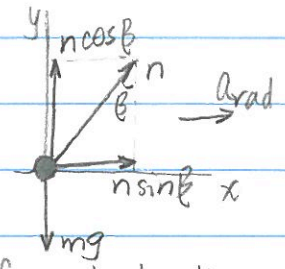
5.115



The bead moves in a circle of radius

$R = r\sin\beta$

The normal force exerted on the bead by the hoop is radially inward



The free-body diagram for the bead

$\Sigma F_y = ma_y \qquad n\cos\beta - mg = 0 \qquad n = \frac{mg}{\cos\beta}$

$\Sigma F_x = ma_x \qquad n\sin\beta = ma_{rad}$

combine these two equations to eliminate n:

$\left(\frac{mg}{\cos\beta}\right)\sin\beta = m\,a_{rad} \qquad \frac{\sin\beta}{\cos\beta} = \frac{a_{rad}}{g}$

$a_{rad} = \frac{v^2}{R}$ and $v = \frac{2\pi R}{T}$, so $a_{rad} = \frac{4\pi^2 R}{T^2}$, where T is the time for one revolution.

$R = r\sin\beta$, so $a_{rad} = \frac{4\pi^2 r\sin\beta}{T^2}$

use this in the above equation: $\frac{\sin\beta}{\cos\beta} = \frac{4\pi^2 r\sin\beta}{T^2 g}$

This equation is satisfied by $\sin\beta = 0$, so $\beta = 0$, or by

$\frac{1}{\cos\beta} = \frac{4\pi^2 r}{T^2 g}$, which gives $\cos\beta = \frac{T^2 g}{4\pi^2 r}$

(5)

(a) 4.00 rev/s implies $T = \left(\frac{1}{4.00}\right)s = 0.250\,s$

Then $\cos\beta = \frac{(0.250\,s)^2(9.80\,m/s^2)}{4\pi^2(0.100\,m)}$ and $\beta = 81.1°$

(b) This would mean $\beta = 90°$. But $\cos 90° = 0$, so this requires $T \to 0$. So β approaches $90°$ as the hoop rotates very fast, but $\beta = 90°$ is not possible.

(c) 1.00 rev/s implies $T = 1.00$ s

The $\cos\beta = \frac{T^2 g}{4\pi^2 r}$ equation then says $\cos\beta = \frac{(1.00\,s)^2 (9.80\,m/s^2)}{4\pi^2 (0.100\,m)} = 2.48$,

which is not possible. The only way to have the $\Sigma \vec{F} = m\vec{a}$ equations satisfied is for $\sin\beta = 0$. This means $\beta = 0$; the bead sits at the bottom of the hoop.